

Project Proposal: Sarcasm Detection with Word Embeddings

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Problem Statement

Sarcasm is a form of figurative language where the literal meaning of words differs from the intended meaning. This makes sarcasm detection particularly challenging for Natural Language Processing (NLP) systems, as traditional sentiment analysis models often misclassify sarcastic text. For example, a sentence like *“Oh great, another rainy day”* appears positive but actually conveys negative sentiment. Misinterpretation of sarcasm can significantly reduce the accuracy of applications such as sentiment analysis, customer feedback analysis, and social media monitoring.

This project aims to build a sarcasm detection model that improves classification accuracy by leveraging pre-trained word embeddings and a deep learning architecture.

Dataset

We will use two publicly available datasets from Kaggle:

- **News Headlines Dataset for Sarcasm Detection**

<https://www.kaggle.com/datasets/rmisra/news-headlines-dataset-for-sarcasm-detection>

This dataset contains news headlines labeled as sarcastic (1) or not sarcastic (0) and is widely used for benchmarking sarcasm detection tasks.

- **Tweets with Sarcasm and Irony Dataset**

<https://www.kaggle.com/datasets/nikhiljohnk/tweets-with-sarcasm-and-irony>

This dataset contains labeled tweets annotated for sarcasm and irony, providing examples from informal, real-world social media contexts.

Using both datasets will allow us to evaluate performance across different text domains: formal headlines versus informal social media posts.

Methodology (Tentative)

Our approach will include the following steps:

1. **Preprocessing:** Tokenize and clean the text, then pad sequences to ensure uniform input length.
2. **Embedding Layer:** Experiment with two types of pre-trained embeddings:

- **Word2Vec** (using Gensim)
 - **GloVe** (Global Vectors for Word Representation)
3. **Model Architecture:** Use a **Bidirectional LSTM (BiLSTM)** to capture context from both directions, followed by dense layers and a sigmoid output for binary classification.
 4. **Training and Evaluation:** Train the model on $\sim 70\%$ of the dataset and evaluate on the remaining $\sim 30\%$, reporting accuracy, precision, recall, and F1-score.
 5. **Comparison:** Compare performance between Word2Vec and GloVe embeddings to determine which yields better results for sarcasm detection.
 6. **Extension (if time permits):** Explore contextual embeddings (e.g., BERT) or attention mechanisms to further improve accuracy.

Conclusion

This project falls under the category of **Comparative Analysis**, as it compares multiple embeddings (Word2Vec vs. GloVe) on the same task. The expected outcome is a robust sarcasm detection model and an evaluation of how different embeddings influence performance across both formal (headlines) and informal (tweets) text.